Fiscal Year:	FY 2019	**Task Last Updated:**	FY 08/05/2019
PI Name:	Nickerson, Cheryl A Ph.D.		
Project Title:	High Dimensional Biology to Understand the Functional Response of Salmonella to Long-Term Multigenerational Growth in the Chronic Stress of Microgravity		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Microbiology		
Space Biology Cross-Element Discipline:	(1) Reproductive Biology		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
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City:	Tempe	**State:**	AZ
Zip Code:	85287-5401	**Congressional District:**	9
Comments:	NOTE PI moved from Tulane University to Arizona State University in 2006.		
Project Type:	Flight	**Solicitation / Funding Source:**	2014 Space Biology Flight (ILSRA) NNH14ZTT002N
Start Date:	06/01/2015	**End Date:**	04/30/2020
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA ARC
Contact Monitor:	Sato, Kevin	**Contact Phone:**	650-604-1104
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Flight Program:	ISS		
Flight Assignment:	ISS NOTE: Extended to 4/30/2020 per NSSC information (Ed., 6/26/18) NOTE: Extended to 5/31/2019 per F. Hernandez/ARC (Ed., 5/8/18)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Abu-Ali, Galeb D.V.M., Ph.D. (Harvard College School of Public Health) Barrila, Jennifer Ph.D. (Arizona State University) Huttenhower, Curtis Ph.D. (Harvard College School of Public Health) Ott, Mark Ph.D. (NASA Johnson Space Center) Travisano, Michael Ph.D. (University of Minnesota)		

Grant/Contract No.:	NNX15AL06G
Performance Goal No.:	
Performance Goal Text:	
Task Description:	While unexpected responses in microorganisms are well documented for short duration culture, the impact of long duration, multigenerational growth of microorganisms in the chronic stress of microgravity has not been investigated. The implications of this gap in knowledge are tremendous, as the response to short duration growth in microgravity alters gene expression, pathogenesis-related stress responses (ex., biofilm formation and resistance to antibiotics), and virulence. Moreover, it is known that stress responses (including chronic stress) can increase mutagenesis in bacteria leading to genomic plasticity, which may provide novel mechanistic insight into genomic instability underlying resistance to antibiotics, pathogenicity, and virulence. We propose that long-term growth in spaceflight culture will induce heritable genomic and epigenetic changes in microorganisms with adverse phenotypic outcomes that could have a profound impact on the risk to crew health and vehicle system performance. Our previous studies aboard the Space Shuttle investigated the response of the model enteric pathogen, Salmonella enterica serovar Typhimurium (S. Typhimurium) to short duration spaceflight culture and demonstrated a significant increase in virulence in a murine model of infection, in addition to biofilm formation, as compared to identically grown cultures on the ground. In addition, global microarray and proteomic analyses revealed that 167 transcripts (including small noncoding RNAs/ncRNAs) and 73 proteins changed expression, with the evolutionarily conserved protein Hfq identified as a likely global regulator involved in the response to this environment. Hfq binds to small ncRNAs thereby facilitating their association with mRNAs, the outcome of which plays a diverse role in regulating gene expression, virulence, stress responses, antibiotic resistance, DNA methylation, and genome stability. We hypothesize that long-term multigenerational culture of S. Typhimurium in the spaceflight environment will result in stress-induced mutations and genomic instability as reflected in changes in the genomic, epigenetic, transcriptomic, and select phenotypic characteristics (including virulence) that will reflect a central role for Hfq as a global regulator of these heritable responses. To address this hypothesis, we propose to characterize the impact of long term spaceflight culture (at least 100 generations) of an S. Typhimurium wild type and hfq mutant on the genomic, epigenetic, transcriptomic, virulence, and pathogenesis-related characteristics of these strains. This proposal addresses Research Emphasis 2 - ISS (International Space Station) Microbial Observatory Studies from section B.3 of this announcement, as it is designed to “take advantage of the technological maturity, low cost, and speed of genomic analyses and the rapid generation time of microbes to monitor the evolution of microbial genomic changes in response to the selective pressures present in the spaceflight environment.” In addition, by employing genomic, transcriptomic, and epigenomic analysis, this proposal also addresses the Research Emphasis 1 – Spaceflight Omics Studies in section B.2. In its design, this proposal will address the NASA Space Biology Plan, Overarching Questions CMM-1a, CMM-1d, and CMM5. This proposal also directly addresses the highest priority recommendations from the Decadal Survey for the establishment of an ISS Microbial Observatory conducting long-term, multigenerational studies of microbial population dynamics (Priority P1). In addition, this study analyzes “microbial growth and physiological responses to the multiple stimuli encountered in spaceflight environments,” as described by the panel in Priority P2.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This research will enrich life on Earth through the use of space technology and the application of biomedical knowledge. Specifically, this study will utilize the microgravity spaceflight platform to 1) to broaden our knowledge of the host-pathogen interaction that leads to infectious disease, and 2) for the development of new therapeutic strategies to combat infectious disease for the general public.
Task Progress:	· Performed a series of science verification tests (SVTs) before experiment verfication test (EVT)-1, with both the wild type and mutant Salmonella strains to confirm successful culture transfers of the targeted volume of bacterial inocula from bag-to-bag. The hardware continued to be updated, optimized, and tested as needed. Worked closely with our Techshot hardware team to confirm fidelity and biocompatibility of the hardware updates, including those involved with the pumps (pump function before and after autoclaving, flow rate and duration to ensure full transfer of the desired bacterial inocula into each bag), biocompatibility of new tubing, and established degassing procedures to minimize the potential for bubble formation in the culture medium during and after loading into flight bags. · Performed EVT-1. Successfully validated science procedures for pre-flight preparation of bacterial inocula, culture media, and fixatives. Successfully loaded bacterial culture media into flight hardware and had success with the first few bags within each cassette. Encountered challenges during EVT-1 with bacterial transfers from bag-to-bag with each transfer. The mutant strain faced the biggest challenge. · Performed a series of post-EVT-1 tests, including live imaging of bacterial growth and inocula mixing tests. These tests confirmed that lack of mixing (and not low oxygen) was the issue with bag-to-bag bacteria transfer, where less and less bacteria were being transferred each time. This was particularly challenging for the mutant as it has low motility. To address this challenge, we successfully established a high fidelity mixing protocol to be performed just prior to transfer of culture to a new bag. · EVT-2 to start in July 2019. Invited Presentations: C. Nickerson: Invited speaker, Synthetic Biology, Engineering, Evolution, and Design (SEED) 2018 meeting Scottsdale, AZ, June 3-7, 2018 Invited speaker, “Microgravity: A Novel Research Platform to Advance Human Health”, Biodesign VIP Summer Salon, Flagstaff, AZ, July 18, 2018 Invited speaker and panelist, “Responsibilities of Authorship”, School of Life Sciences Undergraduate Research (SOLUR), Arizona State University, Sept 7, 2018 Invited speaker, ASM Distinguished Lecturer, Eastern Pennsylvania Branch ASM, Philadelphia, PA, September 24, 2018 Invited panelist and speaker, Leading Women: Biotech and Beyond, Phoenix Convention Center, Phoenix, AZ, October 1, 2018 Invited speaker, University of Louisville, Department of Microbiology and Immunology, Louisville, KY, Oct 4, 2018 Invited speaker, ASM Distinguished Lecturer, Eastern New York Branch ASM, Albany, NY, October 16, 2018

	Invited Speaker, Gastronauts, Duke University, Durham, NC, Feb 5, 2019 Invited speaker, ASM Distinguished Lecturer, Missouri Valley Branch ASM, Omaha, NB, March 15-16, 2019 Invited speaker, 3D Tissue Infection Symposium, Wuerzburg, Germany, April 5-7, 2019 Significant Press Coverage https:// ASU team studying bacteria in space, July 22, 2019. https:// Will Mars missions make humans sick? Here's what we know, December 3, 2018. https:// Farewell flat biology – tackling infectious disease using 3-D tissue engineering, September 10, 2018. https://
Bibliography Type:	Description: (Last Updated: 07/02/2025)
Articles in Peer-reviewed Journals	Barrila J, Crabbé A, Yang J, Franco K, Nydam SD, Forsyth RJ, Davis RR, Gangaraju S, Ott CM, Coyne CB, Bissell MJ, Nickerson CA. "Modeling host-pathogen interactions in the context of the microenvironment: 3-D cell culture comes of age." Infect Immun. 2018 Oct 25;86(11). pii: e00282-18. Print 2018 Nov. Review. https://doi.org/10.1128/IAI.00282-18 ; PubMed PMID: 30181350; PubMed Central PMCID: PMC6204695 , Oct-2018
Awards	Barrila J. "Presidential Early Career Award for Scientists and Engineers (PECASE) presented to Jennifer Barrila, CoInvestigator. She is one of 18 NASA researcher awardees. July 2019." Jul-2019
Books/Book Chapters	Yang J, Thornhill SG, Barrila J, Nickerson CA, Ott CM, McLean RJC. "Microbiology of the built environment in spacecraft used for human flight." in "Microbiology of atypical environments. Methods in Microbiology. Vol. 45." Ed. V. Gurtler, J.T. Trevors. London: Academic Press/Elsevier, 2018. p. 3-26. https://doi.org/10.1016/bs.mim.2018.07.002 , Aug-2018
Books/Book Chapters	Ott CM, Crabbé A, Wilson JW, Barrila J, Castro S, and Nickerson CA. "Microbial Stress: Spaceflight-induced alterations in microbial virulence and infectious disease risks for the crew." in "Stress Challenges and Immunity in Space. From Mechanisms to Monitoring and Preventive Strategies, 2nd Edition." Ed. A. Chouker. Berlin, Heidelberg: Springer, Due: October 18, 2019. In press as of August 2019., Aug-2019